

determined via regression analysis using monthly data over a longer period of time, e.g., a five-year period, with 60 monthly observations.

The cross-sectional model is written as:

$$r_{it} = x_{i1}^* \hat{f}_{1t} + x_{i2}^* \hat{f}_{2t} + \dots + x_{ik}^* \hat{f}_{kt} + e_{it} \quad (6.41)$$

where x_{ij}^* is the normalized factor loading of company i to factor j . For example,

$$x_{kl}^* = \frac{x_{kl} - E(x_k)}{\sigma(x_k)}$$

where $E(x_k)$ is the mean of x_k across all stocks and $\sigma(x_k)$ is the standard deviation of x_k across all stocks.

And unlike the previous models where the factors were known in advance and we estimate the risk sensitivities, here we know the factor loadings (from company data) and we need to estimate the factors.

Statistical Factor Models

Statistical factor models are also referred to as implicit factor models and principal component analysis (PCA). In these models neither the explanatory factors nor sensitivities to these factors are known in advance and they are not readily observed in the market. However, both the statistical factors and sensitivities can be derived from historical data.

There are three common techniques used in statistical factor models: eigenvalue-eigenvector decomposition, singular value decomposition, and factor analysis. Eigenvalue-eigenvector is based on a factoring scheme of the sample covariance matrix and singular value decomposition is based on a factoring scheme of the matrix of returns (see Pearson, 2002). Factor analysis (not to be confused with factor models) is based on a maximum likelihood estimate of the correlations across stocks. In this section we discuss the eigenvalue-eigenvector decomposition technique.

The statistical factor models differs from the previously mentioned models in that analysts estimate both the factors (F_{kS}) and the sensitivities to the factors (b_{ikS}) from a series of historical returns. This model does not make any prior assumptions regarding the appropriate set of explanatory factors or force any preconceived relationship into the model.

This approach is in contrast to the explicit modeling approaches where analysts must specify either a set of explanatory factors or a set of